Elementary generation and a counting bound for primes $1 \pmod 4$

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Abstract. We give a self-contained Euclidean construction of arbitrarily many primes congruent to 1 modulo 4. An explicit, very weak lower bound for their counting function follows from controlling the product of the constructed primes.

Sequence and notation. OEIS A002144 consists of the primes $5, 13, 17, 29, 37, \ldots$ congruent to $1 \pmod 4$. Let $\pi_{1 \pmod 4}(x)$ count such primes at most x.

Theorem 1. *For every real $x \geq 5$,*

$$\pi_{1 \pmod 4}(x) \;\geq\; \lfloor \log_3 (1 + 2 \log_5 x) \rfloor .$$

In particular, OEIS A002144 contains infinitely many distinct primes.

Proof. Set $Q_0 = 1$. Construct primes $q_1, q_2, \ldots$ inductively, with $Q_m = \prod_{j=1}^{m} q_j$ after m steps. Given Q_m, consider

$$N_m = (2Q_m)^2 + 1 = 4Q_m^2 + 1 > 1.$$

If q is any prime factor of N_m, then q is odd and $q \nmid 2Q_m$. Consequently $(2Q_m)^2 \equiv -1 \pmod q$. Fermat's little theorem gives

$$1 \equiv (2Q_m)^{q-1} \equiv (-1)^{(q-1)/2} \pmod q.$$

As q is odd, $(-1)^{(q-1)/2}$ is ± 1, and it must equal 1; hence $q \equiv 1 \pmod 4$. Further, for every previously chosen q_j, $N_m \equiv 1 \pmod{q_j}$. Thus any prime factor q of N_m is a new prime of the required residue class. Choose one and call it q_{m+1}.

The size of our constructed product satisfies

$$Q_{m+1} = Q_m q_{m+1} \leq Q_m (4Q_m^2 + 1) \leq 5Q_m^3 \qquad (Q_m \geq 1).$$

By induction from $Q_0 = 1$,

$$Q_m \leq 5^{(3^m - 1)/2} \quad (m \geq 0).$$

Indeed, replacing m by $m + 1$ multiplies the previous exponent by 3 and adds 1, yielding $(3^{m+1} - 1)/2$. For each integer $m \geq 0$ with $3^m \leq 1 + 2 \log_5 x$, we have $Q_m \leq x$. Since each of the m distinct positive prime factors $q_1, \ldots, q_m$ is at most Q_m, they all lie below x. Taking the largest such integer m proves the claimed lower bound. □

Scope and priority. This is a standard Euclidean-type infinitude argument with a crude explicit quantitative consequence, not a resolution of any open prime-distribution conjecture. The result is classical in spirit; no priority or novelty is asserted.

References.

1. N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, A002144, "Pythagorean primes: primes of the form $4k + 1$," consulted 12 November 2026.
2. G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008; classical elementary proofs concerning primes in arithmetic progressions.

Primary author (AI): ChatGPT (OpenAI). This note was produced with AI assistance. The argument was checked by independent algebraic derivation and computational spot checks; it has not been independently refereed. All statements and errors remain subject to human verification. This note makes no claim of novelty.